



THE SUPPLY SYSTEM

Water, minerals, and food are carried up and down a plant in bundles of tubelike cells. One system, called the xylem, carries water and minerals up the plant. Another, called the phloem, can carry food either upward or downward to whichever parts need it.

Network of veins in a dicot leaf

Midrib



Parallel veins in a monocot leaf

LEAF VEINS

Flowering plants are either monocotyledons (monocots for short) or dicotyledons (dicots). Monocots usually have parallel veins in their adult leaves. The adult leaves of dicots usually have a network of veins.



FLOWER BUD

The flower bud is covered by protective structures known as sepals.

OPENING BUD

As the petals grow, they extend past the sepals and open to reveal the anthers and stigmas (p. 17), which are the male and female parts of the flower.

FLOWER

The brightly colored petals attract insects that will pollinate the flower. The anthers produce pollen, and the stigmas ripen, ready to receive pollen from other flowers (pp. 22–23).



Sepal



Petal



Stigma

Anther



AN OLD PASTIME

Gardening is by no means a new hobby. This 16th-century gardener certainly seems to know what he is doing, even if some of his tools are slightly different from modern ones.



Woodcut by the 16th-century botanist Clusius, showing the tree mallow in its entirety